24EC3801 – EMBEDDED SYSTEMS AND IOT

UNIT I – EMBEDDED COMPUTING

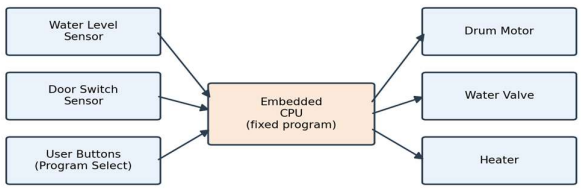
Application/Scenario-Based Question Bank (Apply / Analyze Level)

Course Outcome Mapped: CO1 – Understand and apply the principles of embedded computing systems

Topic Group Distribution (T1–T6)

Group	Syllabus Topic(s) Covered
T1	Embedding Computers
T2	Characteristics of Embedded Computing Applications
T3	Challenges in Embedded Computing System Design
T4	Performance in Embedded Computing
T5	The Embedded System Design Process
T6	Model Train Controller – Conceptual Specification & Detailed Specification

Question Bank (25 Questions, 5 Marks Each)

S.No	Question	Marks	Topic	CO	RBT (K3/K4)	PI Indicator
1	Study the block diagram of a washing machine controller (Figure Q1). Identify which TWO blocks make this system a true 'embedded' computer rather than a general-purpose computer, and justify your choice based on how the CPU interacts with each block. **Figure Q1: Washing Machine Controller Block Diagram** 	5	T1	CO1	K3	PI 1.1
2	A company wants to reuse the same processor board used in a desktop PC to build a cardiac pacemaker. **(a) Identify two reasons (specific to a pacemaker's operating demands) why this reuse is inappropriate.** **(b) Propose one concrete design change to the board that would make it suitable for the pacemaker.**	5	T1	CO1	K4	PI 1.2
3	a) Explain the performance metrics used in embedded computing. (3) b) Summarize the need for power efficiency in embedded systems. (2)	3+2	T1	CO1	K4	PI 1.2
4	Two products use the same 32-bit CPU core: Product A is a smart thermostat that runs one fixed program forever, while Product B is a tablet computer that runs many user-installed apps. Analyze how the fact of 'embedding' changes the software design approach between A and B. List two concrete differences.	5	T1	CO1	K4	PI 1.2

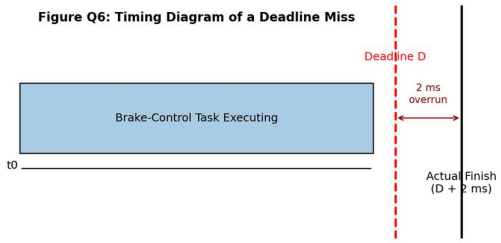
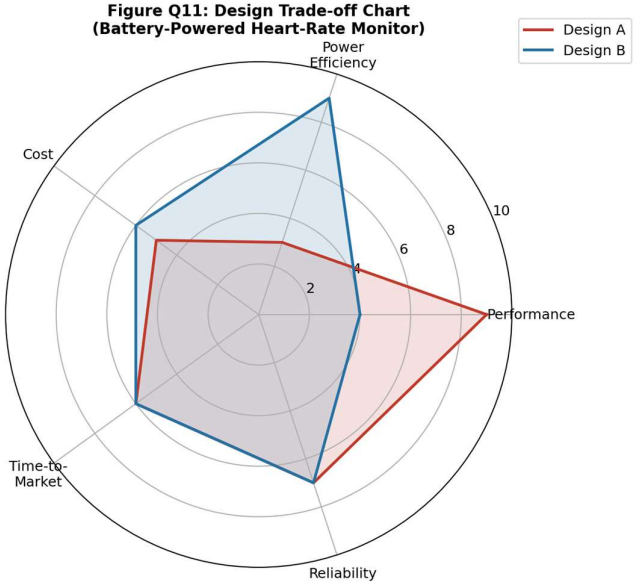
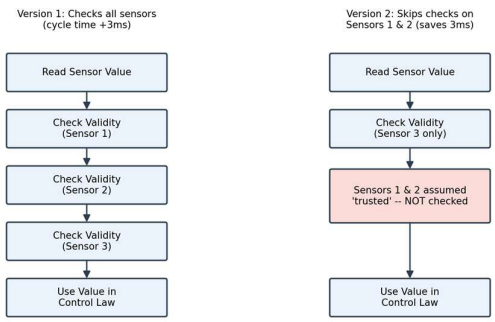
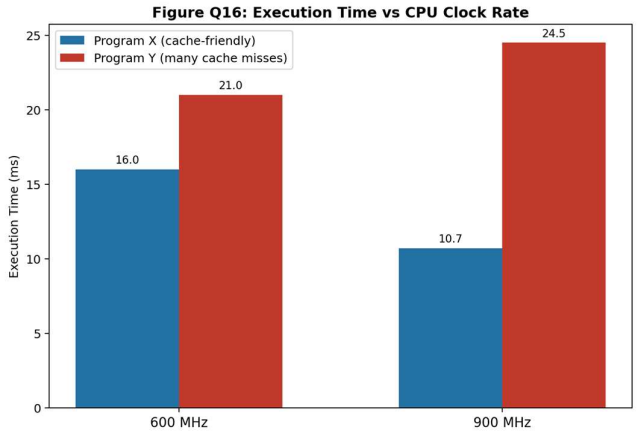
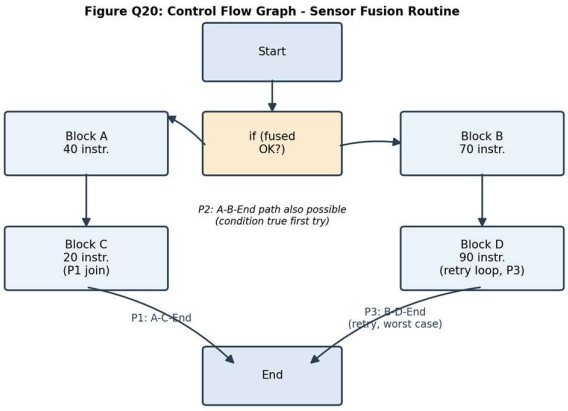
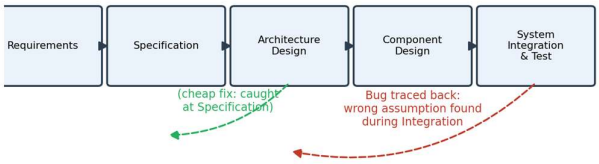
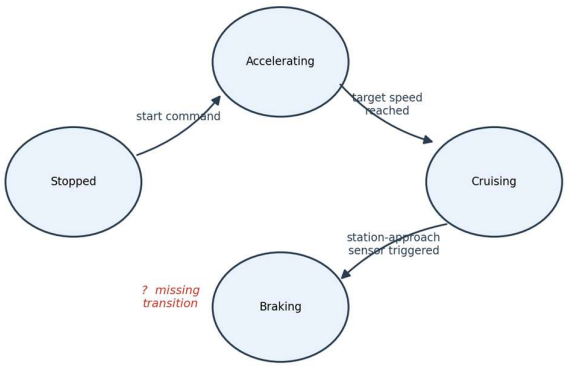
5	A car braking controller is required to complete its control computation before a specified deadline. During testing, the computation occasionally finishes 2 ms after the deadline. Identify the embedded-computing characteristic violated by this behavior and analyze the possible real-world consequence of the deadline violation.	5	T1	CO1	K3	PI 1.1
6	Figure Q6 shows the timing diagram of a car's brake-control task. The task must finish before Deadline D, but it actually finishes 2 ms late. Identify which characteristic of embedded computing applications is violated here, and explain the real-world consequence of this specific 2 ms overrun. **Figure Q6: Timing Diagram of a Deadline Miss**  	5	T2	CO1	K4	PI 1.3
7	A microwave oven's controller must cut power to the magnetron within 50 ms of a 'door open' event, while its display update can tolerate delays of up to 500 ms. **(a) Classify each of these two functions as hard or soft real-time requirements.** **(b) Justify your classification with reference to the consequence of a missed deadline in each case.**	5	T2	CO1	K3	PI 1.1
8	A medical monitor is produced in only 50 units, while a toy controller is produced in 5 million units. The medical monitor requires high reliability, whereas the toy is highly price-sensitive. Analyze how the characteristics of quantity produced, cost and reliability influence the hardware choices for these two embedded products. Give one suitable component-selection decision for each product.	5	T2	CO1	K4	PI 1.3
9	A company can manufacture either 50 units of a special medical monitor (low volume, FDA-regulated, high reliability) or 5 million units of a toy (high volume, price-sensitive). Analyze how the differing 'quantity produced' and 'cost pressure' characteristics should shape the hardware component choices for each product. Give one concrete decision for each product.	5	T2	CO1	K4	PI 1.3
10	A fitness tracker is two weeks away from its committed delivery date. Testing shows that the Bluetooth subsystem consumes 40% more current than the allocated power budget. **(a)** Identify the two design challenges that are directly conflicting. (3) **(b)** Propose one practical design trade-off that could be implemented before delivery and state what it sacrifices. (2)	3+2	T2	CO1	K3	PI 1.1
11	Figure Q11 is a design trade-off chart plotting two candidate designs (A and B) for a battery-powered heart-rate monitor across five metrics. Design A scores high on performance but low on power efficiency; Design B is the reverse. Analyze which design is more appropriate for this specific product, justifying your choice using at least two of the plotted metrics.	5	T3	CO1	K4	PI 1.4

	Figure Q11: Design Trade-off Chart (Battery-Powered Heart-Rate Monitor) 					
12	A design team is two weeks from its committed ship date for a fitness tracker when testing reveals that the Bluetooth module draws 40% more current than the power budget allows. **(a) Identify which two design challenges are in direct conflict in this scenario.** **(b) Propose one design trade-off the team could make in the remaining two weeks, and state clearly what it sacrifices.**	5	T3	CO1	K4	PI 1.4
13	a) Explain the importance of benchmarking and profiling in evaluating embedded system performance. (3) b) Mention any two performance metrics used in embedded computing systems. (2)	3+2	T3	CO1	K4	PI 1.4
14	A wearable ECG device must run for 7 days on a coin-cell battery while sampling continuously at 250 Hz, and it must not miss any abnormal heartbeat event. Analyze the conflict between the 'low power' and 'reliability/correctness' design challenges in this scenario, and suggest one architectural technique that could satisfy both simultaneously.	5	T3	CO1	K4	PI 1.4
15	Figure Q15 shows two candidate flowcharts for a flight-control sensor-input handler. Version 1 checks every sensor value for validity before use; Version 2 skips validity checks on two 'trusted' sensors to save 3 ms per cycle. Analyze which version should be selected for a safety-critical aircraft system, weighing the design challenges involved. **Figure Q15: Flight-Control Sensor-Validation Flowcharts** 	5	T3	CO1	K4	PI 1.4
16	Figure Q16 shows measured execution time for two programs (X: cache-friendly, Y: many cache misses) on two CPUs (600 MHz and 900 MHz). Program X speeds up as expected on the faster CPU, but Program Y actually runs SLOWER on the 900 MHz CPU. Analyze why clock rate alone fails to predict performance here, referring to at least one architectural factor.	5	T4	CO1	K4	PI 1.5

						
17	A digital audio player must decode and output an audio frame every 26 ms to avoid audible glitches. Decoding one frame takes 18 ms on average, but occasionally spikes to 30 ms due to cache misses. **(a) Determine whether this system meets its real-time performance requirement, justifying your answer with reasoning about worst-case vs. average-case execution time. (3)** **(b) Suggest one measure to fix the occasional deadline miss. (2)**	3+2	T4	CO1	K4	PI 1.5
18	a) Explain the embedded system design process. (3) b) State the purpose of conceptual specification in embedded system design. (2)	3+2	T4	CO1	K4	PI 1.5
19	Two embedded processors are considered for a video-decoding task requiring exactly 800 million instructions to decode one frame within a 40 ms deadline: P1 runs at 400 MHz and executes 2 instructions/cycle; P2 runs at 700 MHz and executes 1 instruction/cycle. Determine, with calculations, whether either processor meets the deadline, and recommend the correct choice.	5	T4	CO1	K3	PI 1.1
20	Figure Q20 shows the control flow graph of a sensor-fusion routine with three possible execution paths (P1, P2, P3) and instruction counts per block. Determine the worst-case execution path and its total instruction count, and explain why this measure -- rather than the average case -- is used to set the task's real-time deadline. 	5	T4	CO1	K4	PI 1.5
21	Figure Q21 shows the embedded system design process with feedback paths. A bug discovered during 'System Integration & Test' is traced back to an incorrect assumption made during 'Architecture Design'. Using the diagram, analyze the cost implication of this late discovery compared to if the same bug had	5	T5	CO1	K4	PI 1.6

	been caught during 'Specification', and identify which feedback arrow in the diagram represents each correction. **Figure Q21: Embedded System Design Process with Feedback Paths**  					
22	A team skips writing a formal specification for a smart doorbell and goes straight from requirements to coding. Identify two problems likely to arise later in the project because this step was skipped, and propose what the specification document should have minimally captured to avoid these problems, referencing at least two specification elements (inputs, outputs, functions, performance, cost).	5	T5	CO1	K4	PI 1.6
23	a) Explain the operation of a Model Train Controller using embedded computing principles. (3) b) Mention the function of the Model Train Controller in an embedded system environment. (2)	3+2	T5	CO1	K3	PI 1.1
24	a) Differentiate between conceptual specification and detailed specification. (3) b) State the importance of detailed specification in embedded system development. (2)	3+2	T5	CO1	K4	PI 1.6
25	Figure Q26 shows the state diagram for the model train speed controller (design example), but the transition from 'Braking' back to 'Stopped' is missing its arrow. Using your understanding of the conceptual specification for this design example, state the missing transition together with its triggering condition, and explain why omitting it would leave the specification incomplete. **Figure Q26: Model Train Controller - State Diagram (one transition arrow missing)**  	5	T5	CO1	K4	PI 1.6

Mandatory Validation Summary

Topic Group	No. of Questions	Marks
T1	5	25
T2	5	25
T3	5	25
T4	5	25
T5	5	25
T6	5	25
TOTAL	**30**	**150**